

POCKETSMART AI: SCALABILITY ANALYSIS & FUTURE ROADMAP

1. System Scalability Analysis

PocketSmart AI was engineered with decoupled modular components, allowing the system to scale smoothly from local deployment to enterprise cloud infrastructures:

- **Stateless Asynchronous API Layer:** Built on FastAPI's ASGI architecture, the application handles concurrent asynchronous I/O requests with minimal memory overhead, supporting horizontal scaling behind an Nginx reverse proxy.
- **Decoupled Route Architecture:** Each planner module (/home-budget, /party-budget, /jewelry-budget) functions independently, simplifying future migration to microservice or serverless container environments (e.g., AWS ECS, Google Cloud Run).

2. Database Transition & Data Tier Scaling Roadmap

[Current Architecture]		[Future Enterprise Architecture]
users.json (Flat-File) (Relational Data)	—▶	Amazon Aurora / PostgreSQL
Local Cookie Sessions JWT Token Caching)	—▶	Redis Cluster (Distributed
static/uploads/ (Local FS) Storage Buckets	—▶	AWS S3 / Google Cloud

- **Relational Database Migration:** Transitioning user credentials and recommendation histories from `users.json` to an ACID-compliant PostgreSQL database managed via SQLAlchemy / Alembic migrations.

- **Distributed Cloud Storage:** Relocating local image uploads (static/uploads/) to cloud object storage (AWS S3) with pre-signed upload URLs and CDN caching.

3. Functional & AI Expansion Roadmap

3.1 Live E-Commerce Web Scraping & Affiliate API Integration

Upgrade static search link generation to live API integrations with Amazon Associates, Flipkart Affiliate API, and OYO Partner SDKs to display real-time pricing and stock availability directly within the dashboard.

3.2 Fine-Tuned Open-Source LLM Integration

Introduce self-hosted, quantized open-source models (such as Mistral 7B or Llama 3) running via vLLM alongside Google Gemini, providing cost-effective offline inference for enterprise deployments.

3.3 Multi-Currency Localization & Global Geo-Targeting

Implement automatic currency conversion based on user location, formatting estimates in USD, EUR, GBP, or INR with regional shopping platforms (e.g., Walmart, Target, ASOS).

3.4 Cross-Platform Mobile Application

Develop companion mobile applications for Android and iOS using Flutter, interfacing directly with existing FastAPI REST endpoints.